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Atom Equivalent Energies for the Rapid Estimation of the Heat of Formation of Explosive Molecules from Density Functional Tight Binding Theory

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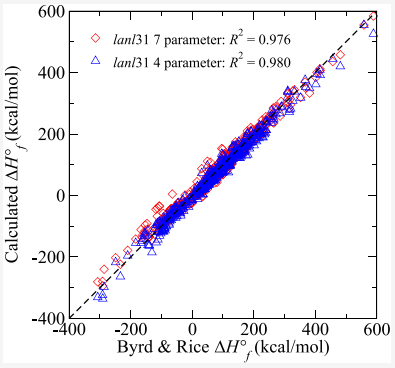
ABSTRACT: Atom equivalent energies have been derived from which the gas-phase heat of formation of explosive molecules can be estimated from fast, semiempirical density functional tight binding total energy calculations. The root-mean-square deviation and maximum deviation of the heats of formation from the experimental values for the set of 45 energetic molecules compiled by Byrd and Rice [*J. Phys. Chem. A*, **2006**, *110*, 1005−1013] are 10.4 and 25.5 kcal/mol, respectively, using 4 atom equivalent energies and 7.4 and 15.0 kcal/mol, respectively, using 7 atom equivalent energies. These errors are around a factor of 2−3 larger than those obtained from density functional theory calculations but are smaller than those obtained from other semiempirical electronic structure methods. Heats of formation calculated with density functional tight binding theory using the 4 and 7 atom equivalent energies, the Byrd and Rice scheme, and the atom pair contribution method for a new set of 531 energetic molecules that contain only carbon, hydrogen, nitrogen, and oxygen are provided.



1. INTRODUCTION

The heat of formation, ΔH_f°, of a substance is the change in enthalpy upon its formation from its constituent elements in their standard states. The heat of formation of explosive molecules and materials is crucial for the estimation of performance metrics, such as detonation velocity and detonation pressure, or impact sensitivity and, as such, is indispensable for the development and characterization of explosives.[1−7] For instance, thermochemical codes such as Cheetah[8] use composition, density, and heat of formation, together with reasonable assumptions about the equation of state of the product species, to estimate detonation properties. While composition and density are relatively easy to determine even for small amounts of explosive materials, accurate measurements for the heats of formation for small samples of sensitive explosives can be difficult. Values for the heat of formation are obtained experimentally by bomb calorimetry,[3] but these experiments require quantities of materials that may not be available during the early stages of explosive synthesis, particularly if the explosive is very sensitive. In the design of next-generation explosives, it will be necessary to be able to accurately predict the heat of formation to screen the performance of new molecules at early stages in their development.

Protocols for the prediction of the heat of formation of gas-phase organic molecules have existed for decades. Benson developed an empirical group additivity scheme, where contributions to the total ΔH_f° are summed over parameterized values for each covalently bonded group of atoms within the molecule.[9,10] This type of approach is dependent on the availability of databases of reliable, experimentally derived heats of formation to which the group contributions are parameterized. In contrast, composite quantum chemical methods such as the Gaussian-2 and Gaussian-3 protocols yield exceptionally accurate atomization energies from which ΔH_f° can be computed to within 1−2 kcal/mol.[11,12] While such *ab initio* methods provide extremely accurate predictions for heats of formation, they are computationally expensive and exhibit an undesirable scaling with the number of atoms that limit their applicability to large molecules.[13]

A family of methods for estimating the gas-phase ΔH_f° of molecules at 298 K has been derived that combine the attractive features of empirical group additivity schemes with the accuracy of more computationally tractable quantum chemical or density functional theory calculations.[14−16] These schemes are based on parameterizing a set of atom or group equivalent energies, ϵ, that map the total energy obtained from an electronic structure calculation, u, to the corresponding heat of formation of the molecule, that is

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Table 1. Training Data Set Used in the Training and Validation of the Atom Equivalent Energies[a]

ID	molecule name	formula	ΔH_f°(kcal/mol) experiment	4-parameter	7-parameter
1	nitromethane	CH_3NO_2	−19.3	−19.79	−24.06
2	tetranitromethane	CN_4O_8	19.7	33.46	26.41
3	azidotrinitro-methane	CN_6O_6	84.2	100.69	92.03
4	*N*-methyl-*N*-nitromethanamine	$C_2H_6N_2O_2$	−1.2	−13.65	−12.83
5	1-azido-1,1-dinitroethane	$C_2H_3NO_4$	60.4	73.08	64.87
6	hexanitroethane	$C_2N_6O_{12}$	42.8	57.97	48.59
7	nitroglycerin	$C_3H_5N_3O_9$	−66.71	−92.19	−79.64
8	hexahydro-1,3,5-trinitroso-1,3,5-triazine	$C_3H_6N_6O_3$	94.3	92.42	106.43
9	cyclotrimethylene trinitramine	$C_3H_6N_6O_6$	45.8	21.6	34.07
10	1,4-dinitroso-piperazine	$C_4H_8N_4O_2$	46.4	50.44	58.83
11	1,4-dinitro-piperazine	$C_4H_8N_4O_4$	13.9	−2.04	5.32
12	*N*-nitrobis(2,2,2-trinitroethyl)-amine	$C_4H_4N_8O_{14}$	21.42	38.6	32.9
13	nitrobenzene	$C_6H_5NO_2$	16.38	10.33	10.64
14	2-nitrophenol	$C_6H_5NO_3$	−31.62	−26.45	−19.6
15	3-nitrophenol	$C_6H_5NO_3$	−26.12	−29.45	−22.59
16	4-nitrophenol	$C_6H_5NO_3$	−27.41	−31.41	−24.56
17	*m*-nitroaniline	$C_6H_6N_2O_2$	14.9	6.8	12.67
18	*p*-nitroaniline	$C_6H_6N_2O_2$	13.2	4.12	9.99
19	azidobenzene	$C_6H_5N_3$	93.0	101.91	100.6
20	1-azido-4-nitrobenzene	$C_6H_4N_4O_2$	93.1	89.94	87.71
21	*N*-nitrobis(2,2-dinitropropyl)amine	$C_6H_{10}N_6O_{10}$	−31.7	−24.34	−29.14
22	4-nitrotoluene	$C_7H_7NO_2$	7.38	2.82	2.66
23	2,4-dinitro-toluene	$C_7H_6N_2O_4$	7.99	−0.78	−1.88
24	benzyl azide	$C_7H_7N_3$	99.5	101.36	99.58
25	3-azido-3-ethylpentane	$C_7H_{15}N_3$	40.6	38.18	29.44
26	2,4,6-trinitrotoluene	$C_7H_5N_3O_6$	5.75	1.11	−0.91
27	2,2-dinitroadamantane	$C_{10}H_{14}N_2O_4$	−36.88	−37.41	−38.31
28	1-azido-adamantane	$C_{10}H_{15}N_3$	51.6	51.32	49.73
29	2-azido-2-phenyl-propane	$C_9H_{11}N_3$	87.4	91.38	88.65
30	2,2′,4,4′,6,6′-hexanitrostilbene	$C_{14}H_6N_6O_{12}$	56.98	49.49	49.67
31	methyl nitrite	CH_3NO_2	−15.64	−9.00	−13.26
32	methyl nitrate	CH_3NO_3	−29.2	−40.52	−38.24
33	dinitromethane	$CH_2N_2O_4$	−14.1	−11.27	−16.47
34	ethyl nitrite	$C_2H_5NO_2$	−25.9	−14.3	−19.04
35	ethyl nitrate	$C_2H_5NO_3$	−37.0	−46.76	−44.96
36	propyl nitrite	$C_3H_7NO_2$	−28.4	−19.59	−24.81
37	*t*-butyl nitrate	$C_4H_9NO_2$	−42.32	−38.98	−44.68
38	*n*-butyl nitrite	$C_4H_9NO_2$	−34.8	−24.62	−30.31
39	2-methyl-2-nitropropane	$C_4H_9NO_2$	−41.0	−25.51	−31.2
40	4,5-furazan-dimethanol dinitrate	$C_4H_4N_4O_7$	2.6	−19.93	−12.41
41	1-nitro-piperidine	$C_5H_{10}NO_2$	−10.6	−20.1	−17.85
42	nitroso-benzene	C_6H_5NO	48.1	48.58	49.41
43	(nitromethyl) benzene	$C_7H_7NO_2$	7.34	8.89	8.72
44	(dinitromethyl)-benzene	$C_7H_6N_2O_4$	8.3	14.79	13.69
45	1,3-dimethyl-2-nitrobenzene	$C_8H_9NO_2$	2.1	−0.47	−1.11
46	methane	CH_4	−17.9	−15.82	−19.15
47	ammonia	NH_3	−11.0	−10.69	−8.0
48	water	H_2O	−57.8	−58.01	−54.33
49	ethyne	C_2H_2	54.2	56.08	56.49
50	ethene	C_2H_4	12.5	12.84	10.39
51	ethane	C_2H_6	−20.1	−21.25	−25.06
52	hydrogen cyanide	HCN	31.5	35.24	34.12
53	carbon monoxide	CO	−26.4	−17.3	−16.18
54	formaldehyde	CH_2O	−26.0	−29.5	−31.24
55	methanol	CH_4O	−48.0	−50.05	−46.84
56	nitrogen dimer	N_2	0.0	4.26	1.61
57	hydrazine	N_2H_4	22.8	11.76	20.02
58	hydrogen peroxide	H_2O_2	−32.5	−31.62	−21.39
59	carbon dioxide	CO_2	−94.1	−96.02	−95.42

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Table 1. continued

ID	molecule name	formula	ΔH_f°(kcal/mol) experiment	4-parameter	7-parameter
60	nitrous oxide	N_2O	19.6	13.56	10.39
61	ozone	O_3	34.1	33.15	31.6
62	propyne	C_3H_4	44.2	45.97	45.91
63	allene	C_3H_4	45.5	46.08	45.27
64	cyclopropene	C_3H_4	66.2	73.74	73.68
65	propylene	C_3H_6	4.8	5.31	2.39
66	cyclopropane	C_3H_6	12.7	12.61	11.18
67	propane	C_3H_8	−25.0	−26.32	−30.61
68	butadiene	C_4H_6	26.3	28.24	26.21
69	2-butyne	C_4H_6	34.8	37.02	36.48
70	methylene cyclopropane	C_4H_6	47.9	47.19	46.65
71	bicyclo-butane	C_4H_6	51.9	63.5	64.46
72	cyclobutene	C_4H_6	37.4	47.28	46.74
73	cyclobutane	C_4H_8	6.8	6.86	4.95
74	isobutene	C_4H_8	−4.0	−1.91	−5.31
75	*trans*-butane	C_4H_{10}	−30.0	−31.44	−36.2
76	isobutane	C_4H_{10}	−32.1	−30.92	−35.68
77	spiropentane	C_5H_8	44.3	47.13	47.6
78	benzene	C_6H_6	19.7	21.65	22.89
79	methyl amine	CH_5N	−5.5	−9.88	−7.66
80	methyl cyanide	C_2H_3N	18.0	22.54	20.95
81	formic acid	CH_2O_2	−90.5	−92.41	−87.6
82	methyl formate	$C_2H_4O_2$	−85.0	−84.53	−80.2
83	acetamide	C_2H_5NO	−57.0	−55.17	−51.83
84	aziridine	C_2H_5N	30.2	30.15	34.75
85	cyanogen	C_2N_2	73.3	67.98	68.6
86	dimethyl-amine	C_2H_7N	−4.4	−8.38	−6.64
87	ethylamine	C_2H_7N	−11.3	−15.16	−13.42
88	ketene	C_2H_2O	−11.4	−11.28	−11.38
89	oxirane	C_2H_4O	−12.6	−15.17	−9.58
90	acetaldehyde	C_2H_4O	−39.7	−38.96	−41.18
91	glyoxal	$C_2H_2O_2$	−50.7	−49.82	−50.44
92	ethanol	C_2H_6O	−56.2	−55.41	−52.69
93	dimethylether	C_2H_6O	−44.0	−41.78	−39.05
94	acrylonitrile	C_3H_3N	43.2	48.08	47.37
95	acetone	C_3H_6O	−51.9	−47.74	−50.43
96	acetic acid	$C_2H_4O_2$	−103.4	−100.69	−96.37
97	isopropanol	C_3H_8O	−65.2	−61.52	−59.27
98	methyl ethylether	C_3H_8O	−51.7	−46.98	−44.73
99	trimethylamine	C_3H_9N	−5.7	−6.2	−4.93
100	furan	C_4H_4O	−8.3	4.06	4.37
101	pyrrole	C_4H_5N	25.9	32.31	30.38
102	pyridine	C_5H_5N	33.6	35.84	35.55
103	hydrogen dimer	H_2	0.0	3.45	0.59

[a]Molecules 1−45 and the corresponding experimental heats of formation were obtained from ref 16, and molecules 46−103 and the corresponding experimental heats of formation were obtained from ref 11. The calculated heats of formation used the optimized atom equivalent energies in Table 2.

$$\Delta H_f^\circ = u - \sum_l n_l \epsilon_l \tag{1}$$

where n_l is the number of atoms or groups of type l, with the equivalent energy ϵ_l in the molecule. The form used in eq 1 is identical to those in refs14−16 but differs trivially from that used by Sattelmeyer et al.[17] by combining the heat of formation of the atoms and electronic energies into ϵ_l. The atom or group equivalent energies, ϵ, are parameterized to experimental measurements of the heat of formation and calculated electronic energies, u, for a set of molecules. The total energies, u, are calculated at the relaxed geometry of each molecule. In this framework, the set of equivalent energies is tied to the level of theory used to compute the total energies in the training step.

High-level quantum chemical or density functional theory calculations with large basis sets can be prohibitively expensive for large molecules, particularly when an iterative geometry optimization is required. Faster computational protocols for obtaining accurate heats of formation would allow us to more rapidly screen large databases of molecules, such as PubChem or GDB-*x*, for potential high-performance explosives or even to develop new molecules *in silico*. We report the development of

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a set of atom equivalent energies that allow the heats of formation of large explosive molecules to be computed from semiempirical density functional tight binding theory[17−21] using the *lanl*31 parameterization[22] with an accuracy that is better than other semiempirical and empirical schemes and at computational expense that is superior to density functional theory. The atom equivalent energies were trained and validated using *k*-fold cross-validation to a set of 103 organic molecules (the "training set"), provided in Table 1, that consisted of the set of 45 energetic molecules compiled by Byrd and Rice[16] and 58 neutral, singlet molecules containing only carbon, hydrogen, nitrogen, and oxygen from the G2/97 data set.[11] We chose to add the 58 small, nonenergetic organic molecules to the training data set to mitigate the potential for overfitting the atom equivalent energies and to assess the transferability of the *lanl*31 parameterization to both energetic molecules and simpler reaction product species such as water, CO, CO_2, and N_2.

We have parameterized two sets of atom equivalent energies. The first uses the same 7 atom equivalent energies employed by Byrd and Rice that distinguishes atoms in single bonds (C, H, N, O) from those in multiple bonds (C′, N′, O′). This formalism is attractive for explosives because the C′, N′, and O′ atom equivalent energies capture the resonant benzene and nitro moieties that are observed frequently in organic explosives. We also parameterized a simpler scheme with 4 equivalent energies for C, H, N, and O that does not account for the bonding environment of the atoms in each molecule. The use of group equivalent energies, such as a contribution specifically for the nitro groups in explosives, can yield smaller statistical errors in the predicted ΔH_f°, but this comes at the expense of having to fit additional parameters, which may require larger sets of training data to reduce uncertainties. For instance, the atom pair contribution (APC) method of Mathieu predicts the heat of formation of gas-phase organic molecules with similar accuracy to our scheme and with significantly faster computational throughput, but it required the parameterization and validation of more adjustable parameters.[23] In contrast, the relatively simple atom equivalent energy scheme developed by Byrd and Rice required experimental data on the gas-phase heat of formation of only 30 explosives to train its seven adjustable parameters.[16]

The 4-parameter atom equivalent energy model is particularly useful for screening large numbers of molecules since only the stoichiometry and total energy, *u*, are required to estimate the gas-phase heat of formation. In contrast, the 7-parameter model can be challenging to automate because atoms must be assigned to the single- or multiple-bonded populations by the analysis of the molecular structure. To apply the 7-parameter model rapidly to large numbers of molecules, we show how bond orders can be extracted directly from the self-consistent density matrix in density functional tight binding theory, which can be used to automatically assign atoms to single- and multiply-bonded environments. Both schemes have been applied to a new data set of 531 explosive molecules (the "application set") for which we provide consistently calculated heats of formation using the 4- and 7-parameter models, the Byrd and Rice density functional theory approach, which uses geometries optimized at the B3LYP/6-31G* level of theory[24] and total energies calculated using the B3LYP exchange−correlation functional[25] but with the larger, 6-311++G(2df,2p) basis set[26,27] without London dispersion corrections, and the empirical APC scheme of Mathieu, along with the corresponding SMILES string for each molecule.

Finally, density functional tight binding has been used successfully by several groups to study shock-induced chemistry in explosives and organic materials[28−35] and to derive Arrhenius chemical kinetics[36,37] for thermal explosions. Hence, the two sets of atom equivalent energies reported for the *lanl*31 parameterization are useful not only because they yield a new scheme for rapidly obtaining heats of formation but also because they allow us to compare, albeit indirectly owing to the variations in how the atom equivalent energies are generated, the accuracy and transferability of some of the different density functional tight binding parameter sets used in reactive molecular dynamics simulations of explosives via estimates of ΔH_f° versus experiment and *ab initio* calculations.[28,29,38] We find that the errors in the ΔH_f° computed using the *lanl*31 parameterization with the atom equivalent energies reported in Table 2 for molecules containing N−O bonds are significantly smaller than those reported by Sattelmeyer et al. using DFTB2/MIO[17,39] and are approximately equal to those reported by Gaus et al. using DFTB3/3OB.[40]

Table 2. Optimized Atom Equivalent Energies for Carbon, Hydrogen, Nitrogen, and Oxygen for the 4- and 7-Parameter Models from the *lanl*31 Density Functional Tight Binding Model and the Corresponding Values for the Byrd and Rice Model that Uses B3LYP/6-31G* Geometries and B3LYP/6-311++G(2df,2p) Energies

	ϵ		
	*lanl*31 4-parameter (kcal/mol)	*lanl*31 7-parameter (kcal/mol)	Byrd and Rice[16] (Hartree)
C	−202.186931	−204.570353	−38.123748
H	−82.884966	−81.455003	−0.597580
N	−187.428944	−194.417592	−54.785466
O	−93.739269	−100.280473	−75.187087
C′		−203.823301	−38.129456
N′		−186.104688	−54.788487
O′		−93.222574	−75.186033

2. METHODS

2.1. Density Functional Tight Binding. Density functional tight binding (DFTB) theory is a semiempirical electronic structure method that is well suited to simulations of organic molecules.[18,19] The framework provides a natural description of the formation of covalent bonds and includes charge transfer between species of different electronegativities. The potential energy, *u*, is

$$u = 2\mathrm{Tr}[(P - P_0)H_0] + \sum_{i=1}^{N} \sum_{j\neq i=1}^{N} \gamma_{ij} q_i q_j + E_{\mathrm{rep}} \tag{2}$$

where H_0 is the charge-independent Slater−Koster tight binding Hamiltonian that represents the distance and angularly dependent overlap between valence orbitals on neighboring atoms,[41] P is the self-consistent density matrix, P_0 is the density matrix for neutral, noninteracting atoms, $\mathrm{Tr}[X]$ denotes the trace of matrix X, γ_{ij} is a screened Coulomb potential,[18] q_i is the Mulliken partial charge on atom i, N is the number of atoms, and E_{rep} is a sum of pair potentials, Φ, that provide strong repulsion at small interatomic distances

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$$E_{\text{rep}} = \frac{1}{2}\sum_{i=1}^{N}\sum_{j\neq i=1}^{N}\Phi(R_{ij}) \tag{3}$$

The self-consistent density matrix, P, is computed from the density functional tight binding Hamiltonian

$$H = H_0 + H_1 \tag{4}$$

and the total number of electrons, where

$$H_1 = \frac{1}{2}S_{i\alpha,j\beta}(V_i + V_j) \tag{5}$$

$S_{i\alpha,j\beta}$ are the elements of the overlap matrix, α and β label orbitals, and

$$V_i = Uq_i + \sum_{j\neq i=1}^{N}\gamma_{ij}q_j \tag{6}$$

is the electrostatic potential arising from the set of atom-centered Mulliken partial charges and the Hubbard U term that describes the change in energy upon charge transfer. The Mulliken partial charges are derived from the density matrix

$$q_i = \sum_{\alpha\in i}(P_{i\alpha,j\beta}S_{j\beta,i\alpha} + P_{j\beta,i\alpha}S_{i\alpha,j\beta}) - \eta_i \tag{7}$$

where η_i is the number of valence electrons on neutral atom i. The electronic energy and interatomic forces in density functional tight binding theory must be calculated self-consistently because the Hamiltonian, H, depends on the density matrix through the Mulliken partial charges.

Density functional tight binding calculations are significantly faster than Kohn−Sham density functional theory calculations because many of the difficult-to-calculate terms in density functional theory are approximated and parameterized in the former. Furthermore, density functional tight binding calculations are typically performed with a minimal basis of atom-centered valence orbitals rather than large basis sets with multiple zeta and polarization functions. The approximations inherent to semiempirical electronic structure models like density functional tight binding theory that minimize their computational expense also give rise to errors. With an optimal parameterization, these errors can be small and well controlled such that accurate predictions can be obtained from simulations.[42] The terms in density functional tight binding models that require parameterization are the radial dependences of the Slater−Koster bond integrals, overlap integrals, and pair potentials, the on-site elements of the Slater−Koster Hamiltonian, which represent the energies of the valence orbitals on neutral, noninteracting atoms, and the values of the Hubbard U's. In this work, we have used the functional forms and parameters that were developed in the *lanl*31 parameterization. This model was demonstrated to exhibit very good transferability to gas-phase organic molecules versus density functional theory, and it provides accurate predictions for the principal Hugoniot loci of organic liquids and molecular crystals versus the experiment.[22]

The density functional tight binding calculations of the energies, u, of the set of molecules given in Table 1 were performed in the gas phase at zero electronic temperature. The electrostatic potentials, V_i, were evaluated in real space. Self-consistency in the Mulliken partial charges was obtained using the DIIS mixing scheme of Pulay with a tolerance of 10^{-6} electrons.[43,44] The geometries of the molecules were optimized using the molecular dynamics-based FIRE algorithm with a tolerance on the maximum force acting on any atom of 10^{-3} eV/Å.[45] All calculations were performed using the open-source code LATTE.[46]

The application of Byrd and Rice's formalism, which distinguishes between atoms that make single or multiple bonds with neighbors, requires a careful analysis of the coordination and valence of each atom. We found that atoms can be assigned rapidly and reliably to the counts of single- and multiple-bonded atoms by the direct evaluation of the total bond order, B_{ij},[47] or Wiberg bond indices,[48] between each pair of atoms i and j from the self-consistent density matrix in the orthogonalized representation $P^{\perp}$

$$B_{ij} = \sum_{\alpha\in i}\sum_{\beta\in j}(2P^{\perp}_{i\alpha,j\beta})^2 \tag{8}$$

where α and β label the orbitals centered on atoms i and j, respectively, and the factor of 2 accounts for the double occupancy of the levels. The total bond order is formally equal to 1 for single bonds, 1.5 for resonant or aromatic bonds, 2 for double bonds, and 3 for triple bonds. We assigned those carbon, nitrogen, and oxygen atoms that form any bond with a total bond order $B_{ij} > 1.25$ to the counts, n_l, of l = C′, N′, or O′, respectively, and to the counts of l = C, N, or O, otherwise. We found that it is particularly convenient to compute the total bond orders from $P^{\perp}$, which is obtained as an intermediate step in the calculation of the DFTB potential energy, u,[49,50] since otherwise both the overlap, S, and density, P, matrices are required.[47] The LATTE code uses the symmetric Löwdin orthogonalization procedure to transform the generalized eigenvalue problem.[51]

The calculations reported here are based on the original self-consistent charge-transfer density functional tight binding scheme of Elstner et al., that is, DFTB2. More advanced density functional tight binding formalisms have been developed, which include, for example, the DFTB3[52] or GFN2-xTB[53] schemes that include additional terms in the expansion of the total energy in charge fluctuations or more recent developments where the sum of pair potentials in the DFTB theory, E_{rep} is replaced by a many-body and/or machine-learned repulsive potential.[54,55] Alternative semi-empirical methods derived from the Hartree−Fock theory rather than density functional theory, such as PM6,[56] also enjoy widespread use for the rapid estimation of heats of formation. While these more advanced schemes may provide more accurate total energies, we concentrate on the *lanl*31 parameterization of the "DFTB2" theory because it remains our workhorse method for reactive molecular dynamics simulations of energetic molecules.

2.2. *k*-Fold Cross-Validation. The two sets of atom equivalent energies, ϵ, were computed by 10-fold cross-validation on the training set of molecules listed in Table 1. Cross-validation methods are particularly attractive in applications where large quantities of training data and separate validation data are unavailable because the model can be trained and validated on the same data set. The training set was split into 10 subsets, each containing 10 or 11 molecules. The energetic and nonenergetic molecules were distributed randomly between the 10 subsets. The atom equivalent energies were parameterized by a linear least-squares fit to nine of the subsets and validated against the remaining subset. This process is repeated so that each subset is used once to validate the values fitted to the other nine

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subsets. The optimal atom equivalent energies obtained from the cross-validation correspond to the average of the k sets of energies, ϵ, obtained from the k optimizations. These values are presented in Table 2. The cross-validation error

$$\mathrm{CV} = \frac{1}{k}\sum_{a=1}^{k}(\mathrm{MSD})_a \tag{9}$$

is equal to the average of the mean-square deviations, MSDs, obtained from the k validation sets. The cross-validation errors obtained from our parameterization procedure were CV = 68.9 and 49.0 $(\mathrm{kcal/mol})^2$ for the models with 4 and 7 parameters, respectively.

3. RESULTS

The transferability of the optimized atom equivalent energies (Table 2) was assessed by comparing the heats of formation computed using eq 1 with the electronic energies, u, calculated using *lanl*31 for the entire training set. The correlation between the calculated and experimental values of the ΔH_f° is depicted in Figure 1a,b for the 4 and 7 atom equivalent energy models, respectively. The root-mean-square and maximum deviations for the training set and the explosive and nonenergetic subsets of molecules are presented in Table 3.

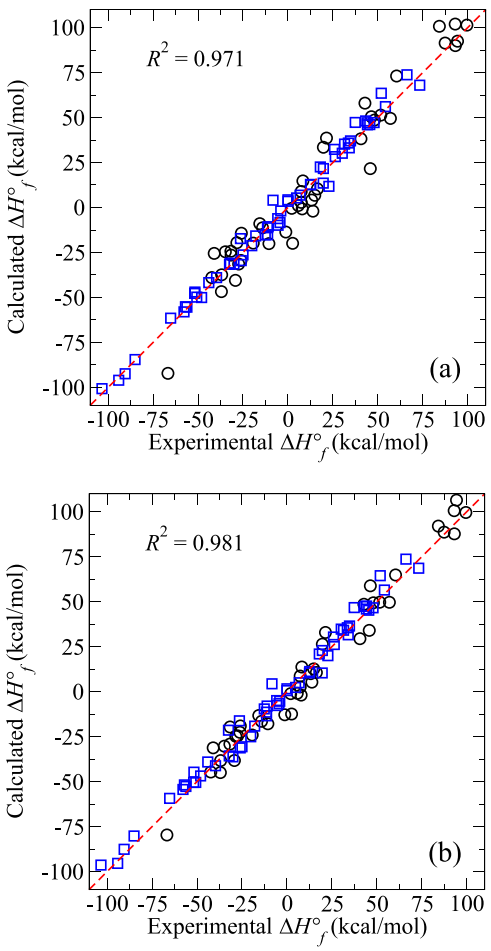

Figure 1. Correlation plot of the heats of formation calculated using the optimized atom equivalent energies and density functional tight binding electronic energies versus experimental data. The 45 energetic molecules and 58 nonenergetic molecules in the training set are represented by black circles and blue square symbols, respectively. The ΔH_f° computed using 4 and 7 atom equivalent energies are presented in (a) and (b), respectively.

The root-mean-square and maximum deviations computed using the 7 atom equivalent energies for the 45 energetic molecules are notably smaller than those obtained using 4 atom equivalent energies. The maximum deviation obtained from the simple model, which uses 4 atom equivalent energies, was 25.5 kcal/mol for the nitroglycerin molecule. It was surprising that the scheme appeared to struggle to give an accurate representation of this molecule because similar molecules were used to train the *lanl*31 model. The model with 7 atom equivalent energies provided a significantly better estimate of the heat of formation of nitroglycerin with a deviation of only 12.1 kcal/mol that we attribute to the ability of the 7-parameter model to distinguish between the single-bonded bridging oxygen atoms from the terminal oxygen atoms in resonance hybrid bonds in the three nitrate ester groups. The maximum deviation obtained from the 7-parameter model was attributed to furazan (15.0 kcal/mol), and the maximum deviation in the subset of 58 nonenergetic molecules came from furan (12.7 kcal/mol). The *lanl*31 model was not trained to either of these heterocycles, and so it is not surprising that the largest errors were obtained for these molecules. In comparison, the maximum deviation reported by Byrd and Rice was only 8.3 kcal/mol.

The additional 3 atom equivalent energies, C′, N′, and O′, in the 7-parameter model, which represent C, N, and O atoms with multiple bonds (resonant, double, triple, etc.), clearly improve the accuracy of the scheme with respect to the simpler, 4-parameter model for the training data. The additional terms are expected to provide a better description of energy of the nitro groups that are common in energetic molecules since the nitrogen and both oxygen atoms are considered to be in multiple-bonded environments. Likewise, the C and C′ terms provide the flexibility for the model to distinguish between explosives based on substituted benzene molecules, such as nitrobenzene, from molecules with saturated carbon backbones, such as nitroglycerin. As a result, the 7-parameter model yields a root-mean-square deviation over the set of 45 energetic molecules of 7.37 kcal/mol, which is 40% smaller than that obtained from the 4 atom equivalent energy model and only a factor of about 2.5 larger than the value of about 3 kcal/mol obtained from the density functional theory calculations of Byrd and Rice. The root-mean-square errors for the subset of 58 nonenergetic molecules are 4.2 and 4.8 kcal/mol for the 4 and 7 atom equivalent energy models, respectively. These values are particularly encouraging because we would expect good-quality density functional theory calculations to give a root-mean-square error of about 3 kcal/mol.[11]

In Figure 2, we present a correlation plot of the ΔH_f° obtained for the training set (Table 1) calculated using the 4- and 7-parameter models. The correlation plot shows clearly that despite the greater accuracy afforded by the 7-parameter model versus the 4-parameter model, we find no systematic difference in the ΔH_f° obtained from the two schemes.

Following the training of the 4- and 7-parameter atom equivalent energy models to experimental data, the ΔH_f° of a new set of 531 explosive molecules have been calculated and the results are compared against predictions from the APC model and the Byrd and Rice density functional theory scheme. The predicted ΔH_f° from the four models are tabulated in the supplied Supporting Information. The density

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Table 3. Root-Mean-Square (RMS) and Maximum Deviations of the Computed Heats of Formation Using the 4- and 7-Parameter Models from Experimental Values for the Training Data and the Subsets of 45 Energetic Molecules and 58 Nonenergetic Molecules

	training data		45 energetic molecules		58 nonenergetic molecules	
	RMS (kcal/mol)	maximum deviation (kcal/mol)	RMS (kcal/mol)	maximum deviation (kcal/mol)	RMS (kcal/mol)	maximum deviation (kcal/mol)
4 parameters	7.59	25.48	10.42	25.48	4.24	12.35
7 parameters	6.08	15.01	7.37	15.01	4.84	12.67

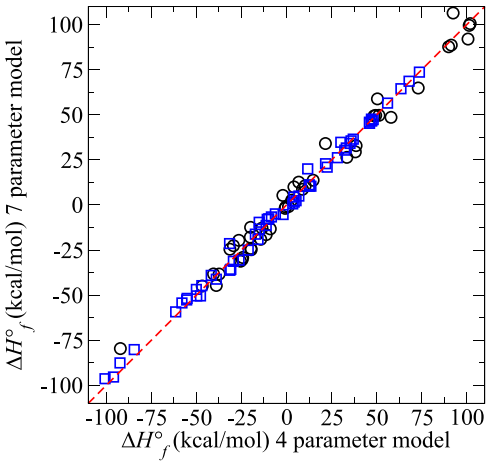

Figure 2. Correlation plot of the heats of formation obtained from the 4- and 7-parameter models for the training set.

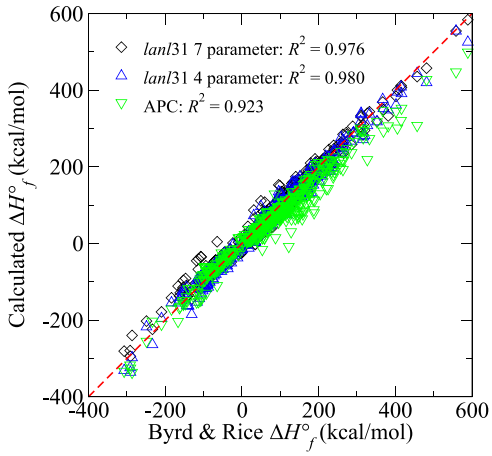

Figure 3. Correlation plot of the predicted heats of formation from the application set from the *lanl*31 density functional tight binding model with 4 and 7 atom equivalent energies and the APC model (527 molecules) versus the Byrd and Rice density functional theory scheme. The coefficient of determination (R^2) for each data set is provided in the legend.

functional theory calculations (B3LYP/6-311++G(2df,2p)//6-31G* energies) for the Byrd and Rice scheme were performed using the Gaussian16 package.[57] An automated workflow was developed to perform the total energy calculations and subsequent analyses in LATTE and Gaussian. We used the Open Babel package[58] to translate the SMILES strings into three-dimensional atomic coordinates that are readable by both codes. The heats of formation of 527 of the 531 molecules in the application set were computed using the APC scheme directly from a list of SMILES strings using the python code provided with ref 23. APC calculations could not be performed on four of the molecules because parameters for N−N triple bonds (N#N) were not available. The accuracy of the heats of formation obtained from density functional tight binding theory and the APC model versus the Byrd and Rice scheme is depicted by correlation plots in Figure 3. The coefficients of determination from the 4- and 7-parameter models and the APC scheme are R^2 = 0.980, 0.976, and 0.923, respectively. The root-mean-square errors from the 4- and 7-parameter models are 16.8 and 18.1 kcal/mol, respectively, whereas the root-mean-square error from the APC model is 30.4 kcal/mol. Hence, both the 4- and 7-parameter atom equivalent energy models based on semiempirical density functional tight binding theory provide, on average, better accuracy than the empirical APC model with respect to the Byrd and Rice scheme, albeit at significantly greater computational expense. Finally, it is interesting to note that while the ability of the 7-parameter model to reproduce the training data is clearly superior to that of the 4-parameter model, both models yield essentially identical errors with respect to the Byrd and Rice scheme when applied to the application set given in the Supporting Information. This suggests that the 7 atom equivalent energies suffer from lower transferability than the simpler 4-parameter scheme, which is likely to be due to the overfitting of the former. These results are also consistent with previous studies that found diminishing returns from increasing the number of atom equivalent energies to distinguish between bonding environments.[59]

4. DISCUSSION AND CONCLUSIONS

The heat of formation of explosive molecules is an important parameter for estimating detonation performance, such as the Chapman−Jouguet pressure, p_{CJ}, and the detonation velocity, D.[60,61] It is therefore important to understand and quantify how errors in the values of ΔH_f estimated by various computational schemes affect predictions of detonation performance. Using the Cheetah package, we have computed p_{CJ} and D for the explosives nitromethane, pentaerythritol tetranitrate (PETN), and cyclotetramethylene tetranitramine (HMX) while varying the ΔH_f by ±10% from their reference values of −27.0, −128.7, and 17.9 kcal/mol, respectively, and keeping the input densities and compositions fixed. The p_{CJ} vary by only 2.5, 0.7, and 0.3% and the detonation velocities by 1.4, 0.4, and 0.06% from their baseline values for nitromethane, PETN, and HMX, respectively. While the characteristic properties that describe detonation performance are evidently relatively insensitive to small variations in ΔH_f, it is nevertheless informative to compare the quality of the predictions of ΔH_f° from density functional tight binding theory using simple atom equivalent energies to those from other fast, semiempirical methods. Elioff et al. used the PM7[62] and RM1[63] schemes to predict the heats of formation of the 45 explosive molecule sets, albeit with several revisions to the reference energies.[64] The root-mean-square deviations from

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Table 4. Heat of Formation for Organic Molecules Containing N−O Double and Resonant Bonds from Experiments and Calculations Using 4 Atom Equivalent Energies with the *lanl*31 and DFTB-MIO[17] Parameterizations[a]

	experiment	*lanl*31		Sattelmeyer et al.[17]	
	ΔH_f° (kcal/mol)	ΔH_f° (kcal/mol)	absolute error (kcal/mol)	ΔH_f° (kcal/mol)	absolute error (kcal/mol)
nitromethane	−17.86	*−19.795*	*1.9*	*−67.097*	*49.2*
nitroethane	−24.38	−27.423	3.0	*−74.650*	*50.3*
1-nitropropane	−29.98	−32.971	3.0	*−80.735*	*50.8*
2-nitropropane	−33.21	−33.001	0.2	*−83.033*	*49.8*
methyl nitrate	−29.11	*−40.516*	*11.4*	*−112.190*	*83.1*
ethyl nitrate	−36.83	*−46.763*	*9.9*	*−120.968*	*84.1*
1-nitrobutane	−34.39	−38.109	3.7	−86.659	52.3
2-nitrobutane	−39.11	−38.109	1.0	−89.455	50.3
nitric acid	−32.1	−45.560	13.5	−114.333	82.2
n-propylnitrate	−41.56	−51.987	10.4	−127.106	85.5
isopropylnitrate	−45.64	−51.931	6.3	−129.838	84.2
nitrous acid	−18.34	−14.794	3.5	−50.966	32.6
methyl nitrite	−15.64	*−8.996*	*6.6*	−48.664	33.0
mean absolute error (kcal/mol)			5.7		60.6

[a]Values in italic font denote that the corresponding molecule was used in the parameterization of the atom equivalent energies.

the PM7 and RM1 calculations are about 20−40% larger than those obtained in this work. The maximum deviations obtained from the predictions at the PM7 and RM1 levels of theory are significantly larger, 53.9 and 58.9 kcal/mol, respectively, than those obtained from the *lanl*31 model (Table 3). While the combination of *lanl*31 density functional tight binding calculations and empirical atom equivalent energies gives, on average, smaller errors than the PM7 and RM1 models, it should be noted that these schemes were not explicitly parameterized to this set of molecules and hence the results are true predictions from the models.

Sattelmeyer et al. computed the heat of formation of 622 organic molecules using the DFTB2/MIO parameterization of density functional tight binding theory with 4 atom equivalent energies.[17] While the accuracy and transferability of the combination of DFTB2/MIO and atom equivalent energies in ref 17 are generally very good, notable exceptions are found for those molecules that contain N−O double or resonant bonds.[39] We present in Table 4 the heat of formation of 13 molecules with the nitro moiety or N−O double bonds calculated using the DFTB2/MIO parameterization from ref 17 and using the *lanl*31 parameterization, both with 4 atom equivalent energies. The mean absolute error in the values of the heat of formation calculated by Sattelmeyer et al., 60.6 kcal/mol, is over 1 order of magnitude greater than those obtained from the *lanl*31 parameterization, 5.7 kcal/mol, with the atom equivalent energies given in Table 2 even though both schemes use the same level of theory. Furthermore, the mean absolute error in the ΔH_f° obtained from the *lanl*31 parameterization at the DFTB2 level of theory for the 13 molecules in Table 4 is only 0.5 kcal/mol larger than that reported using the more sophisticated DFTB3 scheme with the 3OB parameterization with a set atom equivalent energies fitted to the full 622 molecule "Jorgensen" data set.[40]

The root-mean-square deviation for the set of 45 explosive molecules obtained from the empirical APC scheme of Mathieu is 7.36 kcal/mol,[23] which is about 40% smaller than the error obtained from the 4-parameter model and nearly identical to the error obtained from the 7-parameter model, 7.37 kcal/mol. However, the application of the APC model and the 4- and 7-parameter atom equivalent energy models to the application set tabulated in the Supporting Information and depicted in Figure 3 indicates that the transferability of the semiempirical, density functional tight binding scheme to molecules not included in its parameterization is somewhat better than that provided by the APC model.

In summary, two sets of atom equivalent energies for carbon, hydrogen, nitrogen, and oxygen have been derived using *k*-fold cross-validation to experimental values of the gas-phase heat of formation, ΔH_f°, of a set of explosive and nonenergetic molecules. The root-mean-square and maximum absolute deviations in the calculated ΔH_f° are between a factor of 2 and 3 larger than those obtained from density functional theory calculations at the B3LYP/6-311++G(2df,2p) level but smaller than those obtained using other fast, semiempirical models including PM7, RM1, and DFTB2/MIO. Furthermore, both the 4- and 7-parameter schemes developed here exhibit better transferability than a leading empirical model. While the set of 7 atom equivalent energies, which takes into account C, N, and O atoms in different bonding environments, provides better accuracy than the simplest possible model with only 4 atom equivalent energies, the accuracy of the latter is still more than sufficient to screen molecules for thermochemical properties and it does not give rise to systematic errors. The simple scheme requires only a fast calculation of the total energy and the stoichiometry of a molecule to estimate ΔH_f°, which is trivial to implement and automate. The scheme with 7 atom equivalent energies requires a more careful implementation to reliably distinguish whether atoms are in single- or multiple-bonded environments, but this can be accomplished on the fly in density functional tight binding theory by computing the total bond order between each pair of atoms from the density matrix. Hence, we propose that the balance of good accuracy and computation throughput provided by these schemes will be useful in the screening of large databases of organic molecules for explosive performance.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jcim.1c00312.

Supplementary Material (DeltaH_supplement_-March21.xlsx) is provided that lists the SMILES strings,

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counts of atom types for the 4- and 7-parameter schemes, and the heats of formation calculated using the Byrd and Rice density functional theory method, the APC model of Mathieu, and the 4- and 7-parameter atom equivalent energy models with *lanl*31 for the 531 molecule application set presented in Figure 3 (XLSX)

SMILES strings of the molecules listed in Tables 1 and 4 are provided in Molecule_data.xlsx (XLSX)

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Notes

The authors declare no competing financial interest.

All of the density functional tight binding calculations were performed using the open-source code LATTE, which is available for download from https://github.com/lanl/LATTE. The *lanl*31 parameterization is supplied with the LATTE source via the GitHub repository and in ref 22. The APC calculations were performed using the python code provided as the Supplementary Material with ref 23. Density functional theory calculations were performed using Gaussian16, Revision C.01.[57] Open Babel version 3.1.0 was used to convert SMILES to three-dimensional geometries in the standard.xyz format. Thermochemical calculations of detonation performance were performed using Cheetah version 9.0, whose distribution is controlled by the Lawrence Livermore National Laboratory.

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